

Atharv Malusare

AI & Machine Learning Engineer

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EDUCATION

Indiana University Bloomington

Master's in Computer Science (Expected May 2027)

Bloomington, Indiana, USA

Aug 2025 – Present

– Relevant Courses: Software Engineering 1, Applied Algorithms, Applied Machine Learning

Marathwada Mitra Mandal's College of Engineering, SPPU

Bachelor's in Computer Science, CGPA: 8.92 (Honours in Data Science)

Pune, India

Jun 2021 – Jun 2025

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C++, SQL, HTML, CSS, JavaScript
- **Frontend Frameworks:** React, Tailwind CSS (simple, lightweight interfaces)
- **AI/ML Frameworks & Libraries:** PyTorch, TensorFlow, HuggingFace Transformers, LLaMA, OpenCV, YOLO, LangChain
- **NLP & RAG Tools:** Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Vector Databases
- **Data Processing:** Pandas, NumPy, BeautifulSoup, Requests
- **Tools & Platforms:** Git, Docker, AWS (S3, Lambda, EC2), Google Cloud, Selenium

WORK HISTORY

AI Intern

IBN Technology Ltd., Pune, India

Mar 2024 – Jun 2024

- Automated **data ingestion** and preprocessing workflows using **Python** and **SQL**, reducing **data preparation time by 30%**.
- Integrated **ML models** into **server pipelines** for real-time deployment.
- Collaborated with cross-functional teams to identify and implement AI-driven solutions for client projects.

Project Intern

Centre for Development of Advanced Computing (CDAC), Pune, India

Sep 2024 – Jul 2025

- Delivered a **25% boost in answer relevance** over baseline RAG, supporting **20+ languages** and handling **500+ queries/day**.
- Optimized RAG pipelines and LLM integration, reducing **latency from 3.2s to 1.9s**.
- Published a peer-reviewed paper at **ICLI 2025** showcasing chatbot architecture and multilingual performance. [Link]

PROJECTS

AyurSanvaad: Multilingual Ayurvedic Chatbot (Final Year Project) | Python, LangChain, HuggingFace, RAG, AWS, React, Tailwind CSS (Sep 2024 – Jun 2025)

- Built a RAG-based chatbot supporting **20+ Indian languages**, processing **500+ queries/day** with **25% higher relevance** and **40% lower latency**.
- Worked with **CDAC Pune** and consulted by an **Ayurvedic expert** to ensure accurate and well-known responses for the chatbot. Published results in ICLI 2025 [Link].

Helmet & Number Plate Detection | Python, OpenCV, TensorFlow, YOLO (May 2023)

- Built a **real-time detection system** using YOLOv5, achieving **92% mAP** on a custom dataset for **traffic safety compliance**.

Pneumonia & Brain Tumor Detection | Python, CNN, TensorFlow (Dec 2023)

- Developed CNN models achieving **95% accuracy** on test datasets for pneumonia and brain tumor detection, reducing manual diagnostic time by 30%.

Monocular Depth Estimation | Python, MiDaS, Computer Vision (Dec 2023 – Seminar Project)

- Enhanced MiDaS with a **novel attention block**, achieving a **10% reduction in depth error** on indoor scene datasets.
- Tested model on unseen environments to ensure robust generalization.

ACHIEVEMENTS & CERTIFICATIONS

5-star on **HackerRank** in **Python & C++**.

Co-authored and published a **peer-reviewed research paper** at ICLI 2025 on the AyurSanvaad multilingual chatbot [Link].

Machine Learning Specialization (DeepLearning.AI) and **Deep Learning in Computer Vision™**.